Range & Interquartile Range

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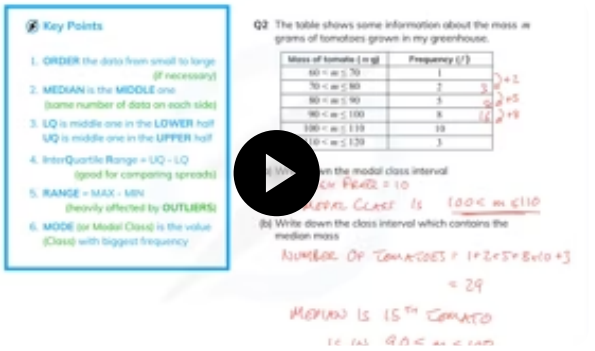

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Range & IQR

What is the range?

- The **range** is the **difference** between the **highest value** and the **lowest value**
  - range = highest - lowest
    - For example, the range of 1, 2, 5, 8 is 8 - 1 = 7
- It measures how **spread out** the data is
  - Ranges of different data sets can be **compared** to see which is more spread out
  - The range of a data set **can be affected** by very large or small values
- Be careful with **negatives**
  - The range of -2, -1, 0, 4 is 4 - (-2) = 6

How do I know when to use the range?

- The **range** is a simple measure of how **spread out** the data is
  - The range does **not** measure an average value
- It should **not** be used if there are any **extreme values** (outliers)
  - For example, the range of 1, 2, 5, 80 is 80 - 1 = 79
    - This is not a good measure of spread

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- The **median** splits the data set into **two parts**
  - Half the data is less than the median
  - Half the data is greater than the median

- **Quartiles** split the data set into **four parts**
  - The **lower quartile (LQ)** lies a **quarter** of the way along the data (when in order)
    - One quarter (25%) of the data is less than the LQ
    - Three quarters (75%) of the data is greater than the LQ

  - The **upper quartile (UQ)** lies **three quarters** of the way along the data (when in order)
    - Three quarters (75%) of the data is less than the UQ
    - One quarter (25%) of the data is greater than the UQ

  - You may come across the median being referred to as the second quartile

How do I find the quartiles?

- Make sure the data is written **in numerical order**
- Use the **median** to divide the data set into **lower and upper halves**
  - If there are an **even** number of data values, then
    - the **first half** of those values are the lower half,
    - and the **second half** are the upper half
    - **All** of the data values are included in one or other of the two halves

  - If there are an **odd** number of data values, then
    - all the values **below** the median are the lower half
    - and all the values **above** the median are the upper half
    - The **median** itself is **not included** as a part of either half

- The **lower quartile** is the **median of the lower half** of the data set
  - and the **upper quartile** is the **median of the upper half** of the data set

- Find the quartiles in the **same way** you would usually find the median
  - just **restrict** your attention to the relevant half of the data

What is the interquartile range (IQR)?

- The **interquartile range (IQR)** is the **difference** between the **upper quartile (UQ)** and the **lower quartile (LQ)**
  - Interquartile range (IQR) = upper quartile (UQ) - lower quartile (LQ)

- The IQR measures how **spread out** the **middle 50%** of the data is
  - The IQR is **not affected by extreme values** in the data

Examiner Tips and Tricks

- If asked to find the range in an exam, make sure you show your subtraction clearly (don't just write down the answer)

Worked Example

Find the range of the data in the table below.

3.4	4.2	2.8	3.6	9.2	3.1	2.9	3.4	3.2
3.5	3.7	3.6	3.2	3.1	2.9	4.1	3.6	3.8
3.4	3.2	4.0	3.7	3.6	2.8	3.9	3.1	3.0

Answer:

Range = highest value - lowest value

9.2 - 2.8

The range is 6.4

Worked Example

A naturalist studying crocodiles has recorded the numbers of eggs found in a random selection of 20 crocodile nests

31 32 35 35 36 37 39 40 42 45

46 48 49 50 51 51 53 54 57 60

Find the interquartile range for this data set.

Answer:

There are 20 data values (an even number)
So the lower half will be the first 10 values
The lower quartile is the median of that lower half of the data

~~31~~ ~~32~~ ~~35~~ ~~35~~ 36 37 ~~39~~ ~~40~~ ~~42~~ ~~45~~

So the lower quartile is midway between 36 and 37 (i.e. 36.5)

Lower quartile = 36.5

Do the same thing with the upper half of the data to find the upper quartile
The upper quartile is the median of the upper half of the data

~~46~~ ~~48~~ ~~49~~ ~~50~~ 51 51 ~~53~~ ~~54~~ ~~57~~ ~~60~~

So the upper quartile is midway between 51 and 51 (i.e. 51)

Upper quartile = 51

Subtract the lower quartile from the upper quartile

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